

- 3 (a) The variation with pressure p of the volume V of a fixed mass of an ideal gas is shown in Fig. 3.1.

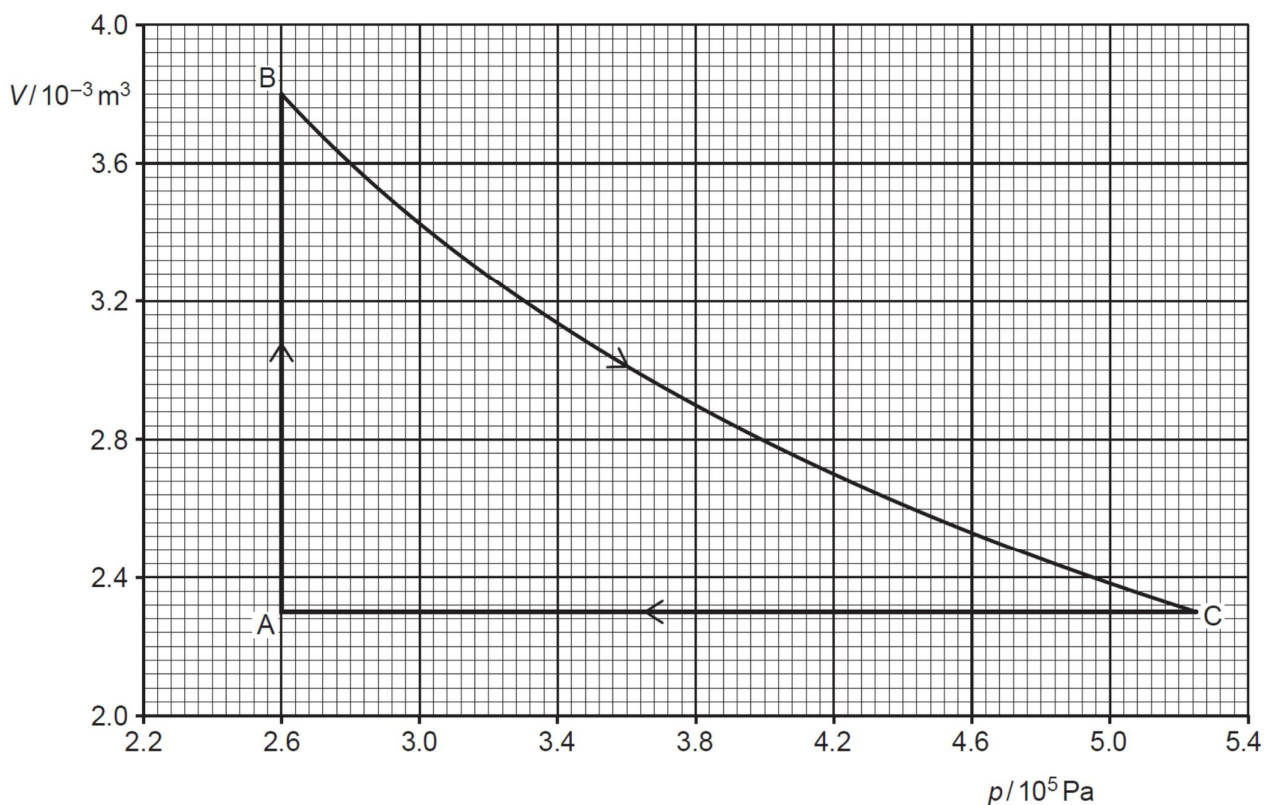


Fig. 3.1

The gas undergoes a cycle of changes A to B to C to A.

During the change A to B, the volume of the gas increases from $2.30 \times 10^{-3} \text{ m}^3$ to $3.80 \times 10^{-3} \text{ m}^3$.

- (i) Show that the magnitude of the work done during the change A to B is 390 J.

[1]

- (ii) State and explain the total change, if any, in the internal energy of the gas during one complete cycle.

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 [2]

- (b) During the change A to B, 1370 J of thermal energy is transferred to the gas.

During the change B to C, no thermal energy enters or leaves the gas. The work done on the gas during this change is 550 J.

Use these data and the information in (a) to complete Table 3.1.

Table 3.1

change	thermal energy supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
A to B			
B to C			
C to A			

[4]

- (c) Use the first law of thermodynamics to explain why the specific latent heat of vaporisation is greater than the specific latent heat of fusion for a particular substance.

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..... [2]

[Total: 9]